



Embeddix Axiom
technologies pvt ltd

— Engineering the Future. Securing Every Mile. —

ENGINEERING SOLUTIONS. DELIVERING REAL IMPACT.

End-to-End Engineering Excellence Across Automotive, Industrial, Consumer Electronics, Electronics, Mechanical & AI/ML Domains.



SAFETY
BY DESIGN



QUALITY &
RELIABILITY



INNOVATION
DRIVEN



PARTNER IN
YOUR SUCCESS



CUSTOMER
CENTRIC



AUTOMOTIVE



INDUSTRIAL



CONSUMER
ELECTRONICS



ELECTRONICS



MECHANICAL



AI / ML



FOCUSED ON
TECHNOLOGY



COMMITTED TO
EXCELLENCE



DRIVEN BY
INNOVATION



DELIVERING
BUSINESS VALUE



www.embeddixaxiom.com

AGENDA



Embeddix Axiom
technologies pvt ltd

Engineering the Future. Securing Every Mile.

A roadmap to explore **who we are**,
what we do and the **value** we deliver.

01

ABOUT US

Know more about
Embeddix Axiom
Technologies.



02

STRENGTH

Our core strengths
that define our
capabilities.



03

COMMITMENT

Our commitment
to quality, safety
and excellence.



04

CAPABILITY OVERVIEW

Comprehensive
overview of our
capabilities.



05

SAFETY LIFECYCLE COVERAGE

End-to-end safety
lifecycle across
product development.



06

SUCCESS STORIES

Real impact delivered
through our
successful projects.



07

END TO END PRODUCT DEVELOPMENT

Complete ownership
from concept to
market.



08

AUTOMATION AI/ML

Leveraging Intelligence
to drive innovation
and efficiency.



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WHO WE ARE...

ENGINEERING SAFETY. DELIVERING RELIABILITY.

Embeddix Axiom is a Technology and Engineering partner delivering **safe**, **reliable** and **innovative** embedded solutions for the automotive industry.

ISO26262 / IEC61508 / ISO21434

Expertise working with standards for Automotive Safety, Cybersecurity and Industrial domain.



Embeddix Axiom
technologies pvt ltd

Engineering the Future. Securing Every Mile.

20+ YRS EXPERIENCE

Two decades of expertise in embedded system design and functional safety domain.

ASIL D/SIL 3 PRODUCTS

Hands-on experience in developing ASIL D / SIL 3 compliant safety products.

CERTIFICATIONS

ISO26262 Functional Safety
EV Core / HIM Safety Assessor
CPRE, Lead - I
Cybersecurity

Embeddix Axiom
technologies pvt ltd

Engineering the Future.
Securing Every Mile.

KNOWLEDGE BASE

Diverse Automotive IPRs in ABS, Ion Braking, Steering, Airbag, ADAS and more.

EMBEDDED PRODUCT DEVELOPMENT

Experience in full product development lifecycle – from concept to end of life support.

TEAM WORK

Skilled, passionate and collaborative team focused on delivering excellence at scale.



SAFETY FIRST
Built with Safety
at the Core



EXCELLENCE DRIVEN
Striving for
Excellence in All



CUSTOMER FOCUSED
Your Success,
Our Commitment



INNOVATION LED
Innovate Today,
Lead Tomorrow



SUSTAINABLE GROWTH
Creating Value for
Long Term



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OUR COMMITMENT

Driven by **Values**. Delivered with **Excellence**.



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— Engineering the Future. Securing Every Mile. —



OUR PROMISE:

Your Vision, Our Commitment – **Together**, We Build What's **Next**.



SAFETY FIRST
Built with Safety
at the Core



EXCELLENCE DRIVEN
Striving for
Excellence in All



TEAMWORK
One Team,
One Goal



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Innovate Today,
Lead Tomorrow



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ELECTRONICS DESIGN CAPABILITY OVERVIEW



Embeddix Axiom
technologies pvt ltd

END-TO-END ENGINEERING. INNOVATIVE SOLUTIONS. RELIABLE RESULTS.

Embeddix Axiom Technologies Pvt Ltd delivers end-to-end electronics design and development services—from concept to end-of-line support—built for performance, safety, and scale.



COMPLETE END-TO-END
From requirements to
end-of-line support



SAFETY & QUALITY FIRST
Compliant designs for
functional safety



FASTER TIME TO MARKET
Optimized processes
and automation



EXPERIENCED TEAM
Domain experts with deep
technical expertise



SCALABLE & FLEXIBLE
Solutions tailored to your
business needs

01 REQUIREMENT ENGINEERING

- Requirement Capture
- Requirement Analysis
- Requirement Traceability
- Requirement Management (using DXL Script)
- Variant Creation (Pure Variant)
- DOORS Setup



02 DESIGN & DEVELOPMENT

- Hardware Design Architecture
- Detailed Design
- Component Selection
- PCB Layout Support (ECAD)
- Design Analysis
 - ✓ Worst Case Analysis
 - ✓ Monte Carlo Analysis
 - ✓ Circuit Simulation



03 FUNCTIONAL SAFETY & ASPICE

- ASPICE Support
- FSM
- System, Hardware, Software Verification Plan
- ISO 26262 (Part 12) Motor Cycle
- Assessment Readiness
- Checklist & Templates



04 VALIDATION & VERIFICATION

- Validation Test Plan as per ISO Standards & OEM Specs
- Data Processing & Engineering Analysis on Validation Data



05 AUTOMATION & PROCESSING

- Tool Development for Automation
- Test Software Development
- Dashboard Development using Power BI
- Data Analytics & Dashboard Development using Python/Dash Library/CI using ASP.NET



06 END-OF-LINE SUPPORT

- Test Coverage Analysis
- Ideas to Improve Test Coverage
- Creation of Manufacturing Test Specification
- Warranty Return Support & Defect Analysis



**WHY PARTNER WITH
EMBEDDIX AXIOM?**



QUALITY ASSURED
Robust processes,
documentation
and reviews



COST OPTIMIZED
Efficient design
and resource
optimization



RISK MITIGATED
Safety-focused
design with thorough
validation



FUTURE READY
Scalable solutions
for evolving market
demands

INDUSTRIES WE SERVE



AUTOMOTIVE



INDUSTRIAL



ENERGY



MEDICAL



CONSUMER
ELECTRONICS



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ECU HW SYSTEM ENGINEERING

ROBUST. COMPLIANT, TRACEABLE.

End-to-end Hardware System Engineering for ECUs with a strong focus on functional safety, traceability, and stakeholder collaboration.



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TOOLS



DOORS



Rhapsody



PTC Integrity



Microsoft
Office



Visio/BI



1. REQUIREMENT ELICITATION (OEM DOCS / CORE)

- Store/Update/Define/Manage OEM reqs. via attributes & Project Document List
- Open Issue List documented and reviewed with stakeholders
- Support CCB where necessary information for Change/Risk assessment.
- Metrics generated and shared with stakeholders



2. HW REQUIREMENT SPECIFICATION (HWRs)

- Analyze and consolidate ECU HWRs requirements and respective attributes.
- Provide traceability via DOORS links to customer/stakeholder requirements.
- Analyze and document verification method and criteria for the requirements.
- Identified review stakeholders and update review attributes
- Schedule joint reviews with stakeholders and manage requirements updates



3. HW ARCHITECTURE DESIGN (HWAD / HSI)

- Analyze and document interface, architecture & functional safety requirements in ECU HWAD
- Creating Hardware and Software interface requirements in HIS/ICD Doors module
- Provide traceability via DOORS links between ECU HWRs to ECU HWAD to HSI
- Analyze and document verification method and criteria for the requirements
- Identified review stakeholders and update review attributes
- Schedule joint reviews with stakeholders and manage requirement updates



4. HW TEST SPECIFICATION (HWTS)

- Consolidate & classify test requirements from existing functional and durability test requirement documents
- Providing traceability between Architecture Design and HW Integration TS, HW Requirements Specifications



**SLIP CONTROL
SYSTEM**

**AIRBAG
SYSTEM**



**FOCUS ON
SAFETY.
QUALITY.
COMPLIANCE.**



STAKEHOLDER COLLABORATION

Structured reviews and
transparent communication
at every stage



END-TO-END TRACEABILITY

DOORS based traceability
across requirements,
design and tests



RISK & CHANGE MANAGEMENT

CCB support, risk assessment
and metrics for proactive
decision making



QUALITY & COMPLIANCE DRIVEN

Aligned with industry standards
and functional safety best
practices



ON-TIME DELIVERY WITH EXCELLENCE

Optimized processes for
predictable and efficient
project execution



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CASE STUDY – PRODUCT DESIGN & DEVELOPMENT

INNOVATIVE DESIGN. RELIABLE PERFORMANCE. DELIVERED.

Embeddix Axiom delivers end-to-end product design and development solutions that drive performance, ensure compliance, and accelerate time to market.



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OVERVIEW

- **Product** : Braking System
- **ASIL Level** : D
- **Client** : Tier 1 Supplier



SCOPE

- Requirement Capture
- HW Architecture, detailed design and component selection
- Design verification and analysis (circuit simulation and testing)
- Functional Safety
- PCB Layout design
- Test Software Development for



CHALLENGES

- Complex design with many actuators and high speed motor controls
- EMI/EMC and Thermal performance
- Reduced BoM cost
- 80% functions and features are ASIL D
- Strict deadlines



SOLUTION

- Team with high domain expertise.
- Design and layout optimization for improved EMI/EMC and Thermal performance
- Low cost component solution
- Ensured seamless coordination between different stakeholders.
- Reviews from SMEs.



VALUE ADD

- Reduced overall timeline by integrated design solution.
- Reduced testing time by the use of test software.
- Established HW design process guidelines



OUTCOME

- Quality design and product
- ASIL D compliant design
- Multiple Successful product launches.



SAFETY BY DESIGN

Functional safety embedded across all stages



QUALITY ASSURED

Robust verification and validation for reliable performance



COST OPTIMIZED

Smart design and component strategy for better value



FASTER TIME TO MARKET

Integrated processes that accelerate delivery



PARTNER OF CHOICE

Trusted by leading OEMs and Tier 1 suppliers



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CAPABILITY OVERVIEW – MECHANICAL DESIGN



Embeddix Axiom
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INNOVATIVE DESIGNS. PRECISE ENGINEERING. RELIABLE PERFORMANCE.

We deliver end-to-end mechanical design solutions from concept to production support, ensuring performance, manufacturability, quality and lifecycle sustainability.



Requirement Analysis



Mechanical Design



Design Analysis



Prototyping



Mechanical & Environmental Testing



Production Support



DESIGN ACTIVITIES



- Requirements analysis
- 3D CAD Parts & Assembly
- Component 2D drawing
- Assembly Drawing
- FEA Report
- MFMEA Report
- Thermal Management



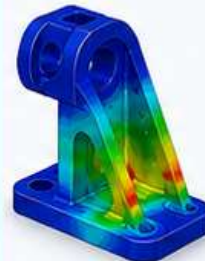
PROCUREMENT SUPPORT & PRODUCT SUSTENANCE



- Mechanical BOM
- Cost analysis
- VAVE
- Obsolescence management
- Product upgrade



SIMULATIONS



- System Engineering
- Concept definition & Simulations
- Dimensional Management
- Structural & Thermal Simulations
- Prototype Development
- Manufacturing Engineering support



TESTING & VALIDATION



- Detailed DVP related to Mechanical design
- Static Verification: Reviews, inspections, and analysis of design documents and models.
- Design Validation and Risk analysis.
- Durability testing

WHY CHOOSE EMBEDDIX AXIOM?



END-TO-END EXPERTISE
From concept to production support



INNOVATIVE & OPTIMIZED DESIGNS
Performance-driven with cost and weight optimization



QUALITY & RELIABILITY ASSURED
Rigorous validation and testing for dependable results



FASTER TIME TO MARKET
Efficient processes and collaboration



SUSTAINABLE SOLUTIONS
Designs for durability, efficiency and environment



TRUSTED BY LEADING OEMs & SUPPLIERS
Proven track record across industries



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ISO 26262 CAPABILITY

FUNCTIONAL SAFETY. SYSTEMATICALLY ENGINEERED.

We follow ISO 26262:2018 standard to deliver safety-critical automotive products with a lifecycle approach—ensuring compliance, reducing risk and safeguarding lives.



ADAS



LANE
DEPARTURE
WARNING

PART 3

CONCEPT PHASE

- Item definition
- HARA
- FSC/FSR



PART 4

SYSTEM DEV. PHASE

- TSC/TSR
- FMEA
- FTA
- Validation Plan
- HSI



PART 5

HW DEV. PHASE

- HSR
- HW Arch & Design Spec
- FMEA
- FTA
- SPFM/LFM
- PMHF
- Verification & Validation Report



PART 6

SW DEV. PHASE

- SSR
- SW Arch & Design Spec
- Verification & Validation report
- SWFM, DFA, FFI



PART 7

PROD & OPERATION

- Production Plan
- User Manual
- Decommissioning Plan



PART 8 & 9 SAFETY ANALYSIS & PROCESS



- ASIL Decomposition
- DFA

- Qualification of HW/SW tools
- CMP
- CiMP



RISK MINIMIZATION

Systematic approach to identify and reduce functional safety risks



COMPLIANCE ASSURED

Aligned with ISO 26262 standard requirements across all lifecycle phases



SAFETY BY DESIGN

Integrated safety in every stage of development from concept to operation



HIGHER RELIABILITY

Robust verification & validation ensures dependable and fault-tolerant systems



FASTED TIME TO MARKET

Proven processes accelerate development with quality and compliance



TRUSTED PARTNER

Experience across ADAS, EV, and safety-critical automotive domains



SAFETY FIRST
Built with Safety at the Core



EXCELLENCE DRIVEN
Striving for Excellence in All



CUSTOMER FOCUSED
Your Success, Our Commitment



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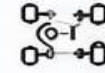


- Overall Safety Management
- Product Safety Management

PART 2 SAFETY MANAGEMENT



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VEHICLE
DYNAMICS
CONTROL



ELECTRIC
VEHICLES



INDUSTRIAL
AUTOMATION

OUR SPECIALIZED SERVICES



TRAINING

- ISO 26262 Awareness to Advanced level training
- Customized training for organization & teams
- Practical hands-on sessions & real-world case studies



PROCESS COMPLIANCE CHECK

- ISO 26262 process compliance assessment
- Gap identification against ISO 26262 requirements
- Recommendations & roadmap to achieve compliance
- Support in process improvement & embedding best practices

CASE STUDY – EV ELECTRIC DRIVE UNIT SYSTEM & COMPONENT LEVEL FUNCTIONAL SAFETY ANALYSIS (ASIL C)



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SCOPE OF WORK

- Safety analysis (System FMEA & SW FMEA) and Quantitative FTA for EV Electric Drive Unit System (Inverter, Electric Motor, Gearbox, Housing).
- Integrated inverter architecture.



SOLUTION

- Used Ansys Medini Analyze for safety architecture integration and analysis.
- Created FMEA structure, mapped functions to system elements, derived malfunctions & linked to hazards.
- Performed FTA for component level failures to derive failure rate and complete quantitative analysis.



VALUE DELIVERY

- Comprehensive safety analysis of complete electric drive unit including inverter, motor control & traceability.
- Variant management by maintaining the core and creating variants with minimal effort.



SAFETY ASSURED

ASIL C compliant design with rigorous safety analysis.



RISK MINIMIZED

FMEA & FTA based approach to reduce functional risks.



TRACEABILITY

End-to-end traceability from safety requirements to component level.



OPTIMIZED VARIANTS

Core maintained with efficient variant management.



RELIABLE & ROBUST

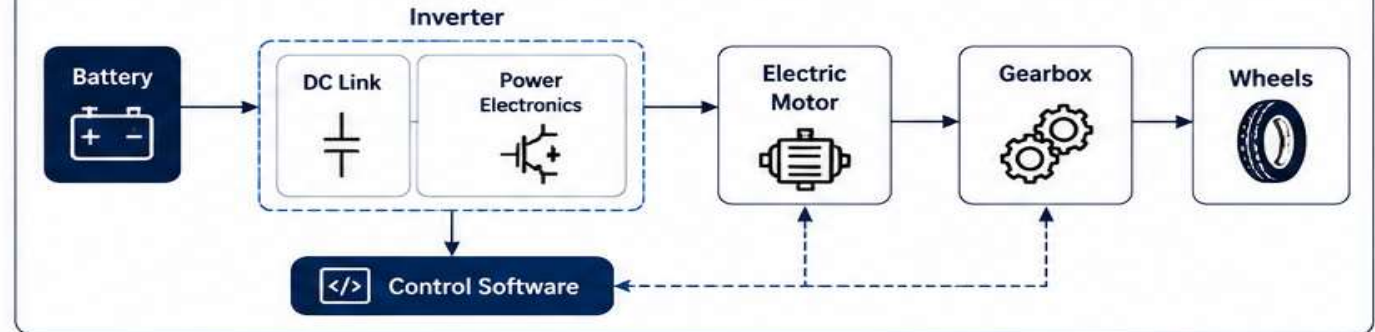
Improved system reliability and performance across operating conditions.



FASTER TIME TO MARKET

Structured process and tools accelerate development and validation.

INTEGRATED ELECTRIC DRIVE UNIT ARCHITECTURE



Inverter: Converts DC from the battery into AC for the motor.



Electric Motor: Transforms electrical energy into mechanical torque.



Control Software: Manages and optimizes system performance.



Gearbox: Adjusts torque and speed for optimal wheel performance.



Housing: Protects components and ensures structural integrity.



LET'S BUILD WHAT'S NEXT. **TOGETHER.**



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END-TO-END DEVELOPMENT OF 2 WHEELER 2 CHANNEL ABS SYSTEM

From Concept to Road – Complete Product Creation with Functional Safety



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Requirements
& System Engineering



Hardware Design
& ECAD / PCB Layout



AUTOSAR Software
Development



Prototype Build



Testing &
Validation



Road Testing



Calibration &
Fine Tuning



Production
Readiness

2 Channel ABS for 2 Wheelers – Designed. Developed. Validated.



OUR CAPABILITIES



Requirement Development

- System, HW & SW requirements with safety goals
- Compliance to ISO26262 & GB standards



Hardware Design

- Schematic, component selection, DFMEA
- ECAD, PCB layout, DFM/DFT



AUTOSAR Software

- MCAL, BSW & ASW development
- Control algorithms with real-time performance



Prototype Manufacturing

- Build, bring-up & assurance testing



Testing & Validation

- HiL, SiL, DVP, EMC/EMI, environmental tests
- ISO26262 V&V work products



Road Testing

- Real world performance validation



Calibration & Fine Tuning

- Slip control, braking performance optimization
- Multiple vehicle/tyre/road conditions



SYSTEM HIGHLIGHTS



Cornering ABS
for Enhanced Safety



Prevents
Wheel Lock



Stable Braking on
All Road Conditions



Better Vehicle
Control



STANDARDS COMPLIANCE



ISO 26262
Functional Safety



GB Standard
Compliant



VALUE PROPOSITION



One-Stop Partner

Complete ownership from concept
to production readiness



Faster Time to Market

Integrated HW + SW + Validation



Proven Engineering Excellence

Domain expertise in control,
functional safety & embedded systems



Reliable & Scalable Solutions

Ready for next-gen 2W mobility
and global benchmarks



End-to-End
Development



Safety
First Design



Optimized
Braking Performance



High Quality
& Robust Design



Future Ready for
BS/EV Mobility

**Engineering Safer, Smarter
and Reliable Mobility**



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INNOVATING SAFETY. ENGINEERING EXCELLENCE.

End-to-End Development of Advanced Industrial Safety Solutions



OUR INDUSTRIAL PRODUCTS



USS JAMMER

Protecting What Matters.
Preventing Unwanted Intrusion.

INDUSTRIAL PRODUCT

- ✓ Blocks unauthorized ultrasonic interference
- ✓ Enhances equipment security and anti-theft protection
- ✓ Real-time detection and adaptive jamming
- ✓ Compact, robust and industrial grade design



USS PROXIMITY SWITCH

Intelligent Control.
Seamless Switching.

INDUSTRIAL PRODUCT

- ✓ Smart control of proximity sensors (ON/OFF)
- ✓ Improves reliability and reduces false triggering
- ✓ System health monitoring and diagnostics
- ✓ ISO 26262 aligned design approach
- ✓ Compact, rugged and industrial grade design

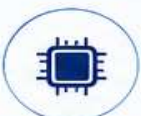


END-TO-END DEVELOPMENT CAPABILITIES



1. REQUIREMENT DEVELOPMENT

Understanding needs, defining requirements and system architecture



2. HARDWARE DESIGN

Schematic design, component selection, and system design



3. ECAD / PCB LAYOUT

High-speed PCB layout
DFM/DFT compliant design



4. SOFTWARE DEVELOPMENT

Embedded software, drivers, algorithms and diagnostics



5. PROTOTYPE MANUFACTURING

Prototype build, assembly and bring-up



6. TESTING & VALIDATION

Functional, integration, HIL and system level testing



7. CALIBRATION & FINE TUNING

Parameter calibration, performance optimization and fine tuning



8. RELEASE & SUPPORT

Production release support and field performance monitoring



SAFETY FIRST

Designed for security and functional safety



QUALITY ASSURED

Built to meet industrial standards & reliability requirements



PERFORMANCE DRIVEN

Optimized for accuracy, robustness and efficiency



INNOVATION LED

Smart solutions for the evolving industrial landscape



PARTNER OF CHOICE

End-to-end ownership with commitment to success

— Safety assessment, gap analysis and process improvement as per IEC 61508 for Baler ECU – UK Client —



INDUSTRY

Industrial Automation



APPLICATION

Baler Machine



PRODUCT

Baler ECU



STANDARD

IEC 61508
Functional Safety



THE CHALLENGE

The client needed an independent safety assessment of their Baler ECU development and associated processes to ensure compliance with IEC 61508 functional safety standard.

- ✓ Uncertainty in achieving the required SIL level (SIL 2) for safety functions
- ✓ Gaps in safety lifecycle activities and work products
- ✓ Inconsistent application of IEC 61508 requirements across development phases
- ✓ Need for actionable recommendations to strengthen the safety process and documentation
- ✓ Preparation for third-party certification and market acceptance in the UK



OUR SOLUTION

Embeddix Axiom conducted a comprehensive safety assessment, identified the gaps and provided clear recommendations to achieve and demonstrate IEC 61508 compliance.

- ✓ Complete safety lifecycle assessment as per IEC 61508:2010
- ✓ Gap analysis against IEC 61508 requirements for SIL 2
- ✓ Evaluation of HARA, SRS, Architecture, HW/SW design, verification & validation, and supporting processes
- ✓ Identification of non-compliances and improvement opportunities
- ✓ Detailed recommendations with practical guidance to close gaps
- ✓ Support for process improvement and work product enhancement



OUR APPROACH



1 PLANNING & SCOPE DEFINITION

Understood system, safety functions, SIL targets and project context



2 DOCUMENT & EVIDENCE REVIEW

Reviewed all safety lifecycle activities, work products and process documentation



3 GAP ANALYSIS

Assessed compliance against IEC 61508 clauses and identified gaps & deficiencies



4 RECOMMENDATIONS & IMPROVEMENT PLAN

Provided actionable recommendations and prioritized improvement plan



5 REPORTING & KNOWLEDGE TRANSFER

Delivered assessment report, discussed findings and supported knowledge transfer



KEY DELIVERABLES

- ✓ Safety assessment report with compliance status
- ✓ Detailed gap list mapped to IEC 61508 clauses
- ✓ Risk & SIL verification findings
- ✓ Recommendations with implementation guidance
- ✓ Process improvement suggestions
- ✓ Enhanced work products and templates
- ✓ Final compliance improvement roadmap



BUSINESS OUTCOMES

- ✓ IEC 61508 compliant development strengthened for SIL 2
- ✓ Improved functional safety assurance and product reliability
- ✓ Reduced risk of non-compliance and certification delays
- ✓ Stronger development process and quality of work products
- ✓ Clear roadmap for continuous improvement
- ✓ Enhanced customer confidence and market acceptance in UK



OUR COMPETENCIES

- ✓ IEC 61508 Functional Safety Expertise
- ✓ Safety Lifecycle Assessment & Gap Analysis
- ✓ Risk Analysis (HARA, SIL Determination)
- ✓ Process Improvement & Compliance Advisory
- ✓ Industrial Embedded Systems Experience



SAFETY FIRST
Built with Safety
at the Core



EXCELLENCE DRIVEN
Striving for
Excellence in All



CUSTOMER FOCUSED
Your Success,
Our Commitment



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Innovate Today,
Lead Tomorrow



SUSTAINABLE GROWTH
Creating Value for
Long Term

CASE STUDY – PROCESS SUPPORT IN ASPICE & ISO26262



Embeddix Axiom
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Driving compliant, efficient and high-quality hardware engineering processes aligned with ASPICE 4.0 and ISO26262 safety standards.



SCOPE OF WORK



Establish hardware engineering process, meet the requirements of ASPICE 4.0 process groups and ISO26262 safety standards.



SOLUTION



Due Diligence Phase

Engaged with customer teams to understand existing processes and perform detail gap analysis with respect to process models. Provide a gap analysis report identifying the process areas and a plan to document the tailored process.



Gap Closure Phase

Prepare tailored process documents, work instructions, templates and checklist documents compliant to ASPICE engineering process groups and ISO26262 functional safety standard.



VALUE DELIVERY



Process Gap
Analysis Report



Process
Documents



Work
Instructions



Templates and
Checklists



COMPLIANCE ASSURED

Aligned with ASPICE 4.0
and ISO26262 standards



STRUCTURED & REPEATABLE

Well-defined processes for
consistent execution



RISK REDUCTION

Proactive gap management
reduces risks and rework



AUDIT & ASSESSMENT READY

Complete documentation for
smooth assessments



EFFICIENCY & QUALITY

Improved productivity
and higher quality outcomes

OUR PROCESS APPROACH



Due Diligence

Understand current processes
and collect information

01



Gap Analysis

Identify gaps against ASPICE 4.0
and ISO26262 standards

02



Gap Closure

Develop and update documents
to close identified gaps

03



Final Readiness Review

Ensure compliance and readiness
for assessment

04



Interim Review

Review progress and
provide feedback



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CASE STUDY – AUTOMATE RE METRICS REPORT PUBLISHING



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Automated, accurate and on-time RE metrics reporting for better decisions.



SCOPE OF WORK

- Association of software quality and maturity metrics variants for V-model, Gov. V, Dev > Mfg as per SPICE models.
- Integrated IBM DOORS with Power BI for real-time project visibility.
- File schedule based reporting for organizational, project wise and dashboard publishing.



SOLUTION

- DOORS Scripting (DOORS automation) weekly export of requirement metrics (.CSV).
- Power BI Desktop consumes .CSV data and drives interactive dashboards.
- OneDrive ensures secure, synchronized storage, enabling reliable access.



VALUE DELIVERY

- Significant improvement of Requirement Engineering metrics reporting.
- Real-time insights into requirement completeness, traceability and maturity.
- Reduced manual review and improved report accuracy.
- Enhanced governance and SPICE compliance with easy management.



AUTOMATED & EFFICIENT

Eliminates manual effort and ensures on-time reporting



REAL-TIME VISIBILITY

Up-to-date metrics for better decision making



ACCURATE & RELIABLE

Improved data quality and report accuracy



GOVERNANCE ASSURED

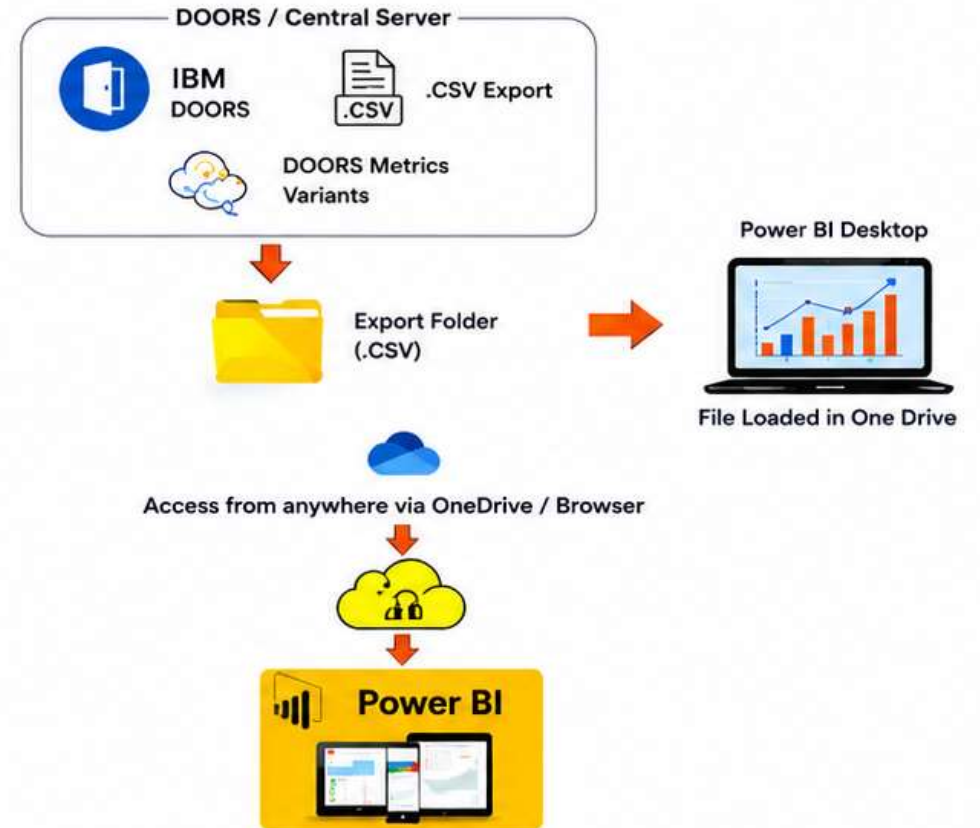
Better compliance with SPICE standards



SCALABLE & SECURE

Cloud-based access with secure and scalable reporting

AUTOMATED RE METRICS REPORTING WORKFLOW



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CASE STUDY – CENTRAL PORTAL OF VALIDATION METRICS

Real-time visibility. Centralized insights. Smarter decisions.



Embeddix Axiom
technologies pvt ltd



SCOPE OF WORK

Digitized validation metrics lifecycle by onboarding data from multiple sources, processing through a central platform, and presenting real-time insights for ISO, ASPICE compliance, project tracking, and trend analysis.



SOLUTION

- **Kafka-based pipeline** aggregates metrics from diversified sources.
- **Automated ETL** to clean, validate, and standardize data in real-time.
- **Secure & scalable storage** ensuring data availability and reliability.
- **Role-based access** for team, project, and system analytics.
- **Intuitive dashboards** for actionable insights and decision-making.



VALUE DELIVERY

- ✓ **Centralized & scalable** data processing
- ✓ **Single source of truth** for metrics
- ✓ **Real-time dashboards & reporting**



REAL-TIME VISIBILITY

Up-to-date metrics for faster decisions



CENTRALIZED & SCALABLE

Single platform for all validation metrics



SECURE & RELIABLE

Role-based access with secure data pipeline



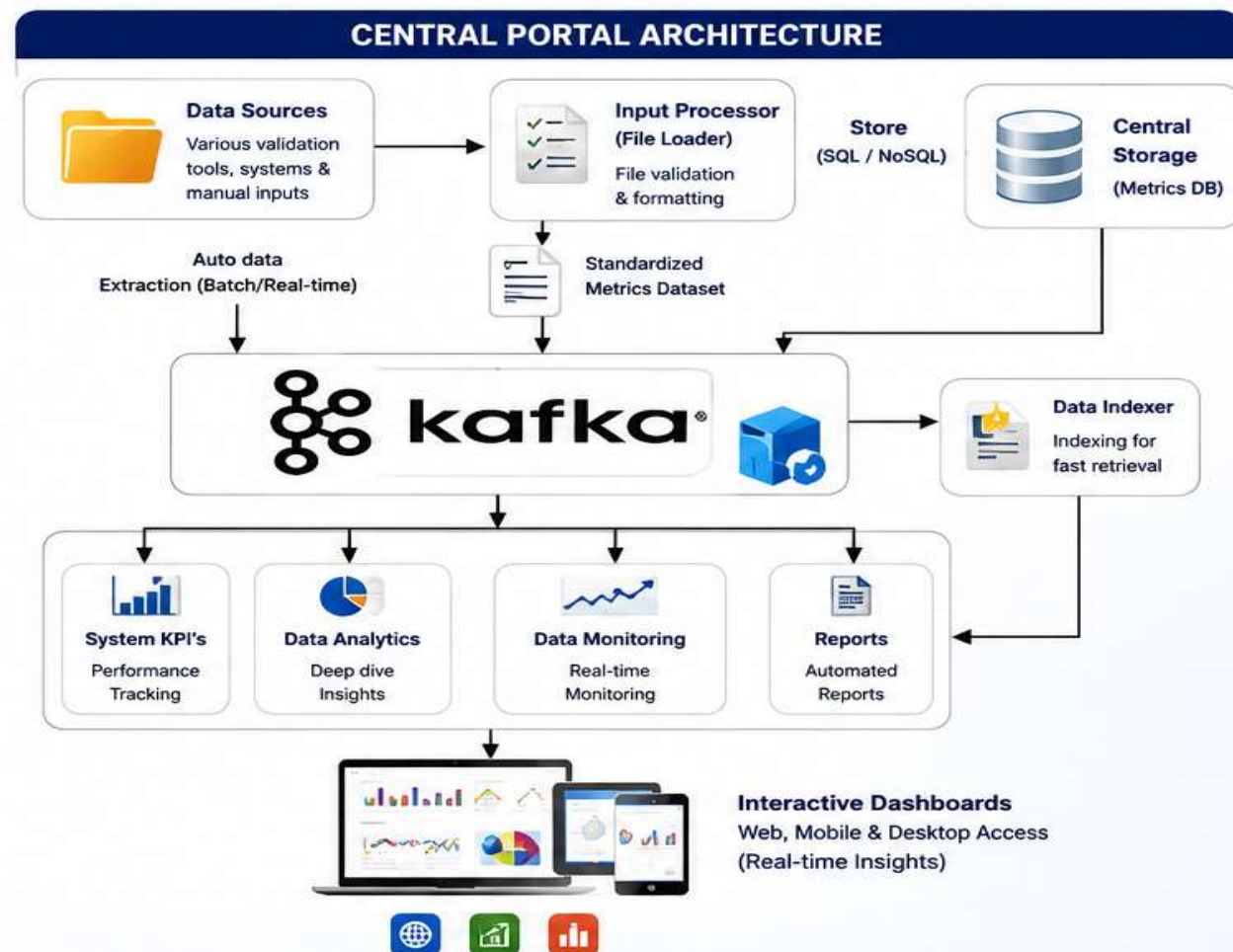
COMPLIANCE ASSURED

Support for ISO, ASPICE & internal quality standards



ACTIONABLE INSIGHTS

Trends, KPIs & reports for continuous improvement



LET'S BUILD WHAT'S NEXT. **TOGETHER.**



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CASE STUDY – PREDICTING SOLAR PV DEGRADATION & LOSSES USING ML

AI/ML driven predictions for higher accuracy, lower losses & better ROI.



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Engineering the Future. Securing Every Mile.



SCOPE OF WORK

- Developed a robust ML framework to estimate and predict photovoltaic (PV) system losses and degradation using historical and real-time data. The model considers weather conditions, soiling, inverter losses, shading, and system configurations to deliver accurate forecasts and optimize asset performance.



SOLUTION

- End-to-end ML pipeline for data collection, preprocessing, feature engineering, model training & validation.
- Predictive models for energy generation, degradation rate, and loss estimation.
- Training with diverse data sources (SCADA, weather, irradiance, soiling, temperature, inverter, shading).
- Actionable insights with dashboards & alerts for performance monitoring and proactive maintenance.



VALUE DELIVERY

- ✓ Accurate prediction of solar generation & PV losses.
- ✓ Early detection of degradation & potential issues.
- ✓ Optimized performance, improved ROI & O&M planning.
- ✓ Data-driven decisions for capacity planning & risk mitigation.
- ✓ Scalable solution adaptable to multiple sites & climates.



HIGHER ACCURACY

Improved prediction accuracy with ML & transfer learning



REDUCED LOSSES

Early detection & proactive actions reduce losses



BETTER ROI

Optimized performance boosts energy yield & ROI



DATA-DRIVEN INSIGHTS

Real-time dashboards & alerts for smarter decisions

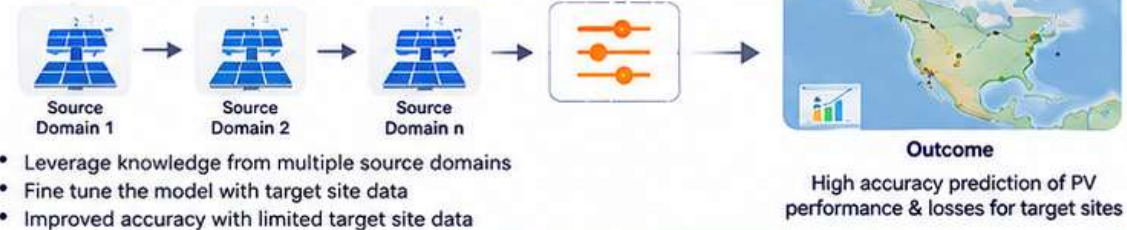


SCALABLE & ADAPTABLE

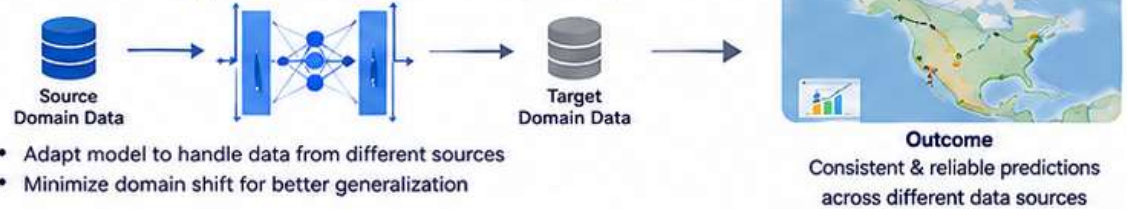
Works across multiple sites, climates & data sources

PREDICTIVE MODELS & OUTCOMES

1. Transfer Learning (Fine Tuning)



2. Domain Adaptation (Data Source Domain)



3. Domain Generalization (Multi Source Domain)



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CASE STUDY

AUTOMOTIVE CYBER SECURITY SUPPORT

Strengthening Vehicle Security. Enabling Trust. Protecting the Future.



CLIENT

Leading Global
Automotive OEM



ENGAGEMENT TYPE

Cyber Security Engineering
& Validation Support



DURATION

18+ Months
(Ongoing Engagement)



Embeddix Axiom has been our trusted cybersecurity engineering partner, providing end-to-end support across the product development lifecycle. Their expertise, ownership and proactive approach have significantly strengthened our product security posture."

BUSINESS CHALLENGE



- Rising cyber threats targeting connected vehicles.
- Compliance with ISO/SAE 21434 and UNECE R155.
- Need for secure by design approach across ECUs.
- Lack of in-house bandwidth and domain expertise.
- End-to-end visibility of cybersecurity in SDLC.

OUR SOLUTION



- Provided end-to-end automotive cybersecurity engineering support.
- Implemented security process aligned with ISO/SAE 21434.
- Performed risk assessment, threat modeling and security validations.
- Supported security architecture review, secure coding & penetration testing.

SCOPE OF SUPPORT



- Cyber Security Management System Implementation
- TARA & Risk Assessment
- Security Architecture Review
- Secure Communication (Crypto, Key Management)
- Secure Boot & Secure Update
- ECU Hardening & Configuration Review
- Static & Dynamic Code Analysis
- Penetration Testing
- Vulnerability Assessment
- Compliance & Documentation (ISO/SAE 21434, UNECE R155)

OUR APPROACH – END TO END CYBER SECURITY SUPPORT



1. REQUIREMENTS & STAKEHOLDER ANALYSIS

Understand asset, business and security goals



2. RISK ASSESSMENT & TARA

Identify threats, vulnerabilities & impact



3. SECURITY ARCHITECTURE & DESIGN REVIEW

Define security controls and mitigation strategy



4. IMPLEMENTATION SUPPORT

Secure coding, configurations, crypto & key management



5. TESTING & VALIDATION

VAPT, fuzzing, code analysis, security test cases



6. COMPLIANCE & DOCUMENTATION

ISO/SAE 21434 work products, audit & regulatory support



7. POST RELEASE SUPPORT

Vulnerability monitoring, incident response support

BUSINESS IMPACT

- ✓ Stronger protection against evolving cyber threats
- ✓ Regulatory compliance (ISO/SAE 21434, UNECE R155)
- ✓ Reduced security vulnerabilities and risk exposure
- ✓ Improved product reliability and customer trust
- ✓ Faster time-to-market with secure by design approach



TECHNOLOGIES & STANDARDS



ISO/SAE 21434



UNECE R155

AUTOSAR

AUTOSAR



HSM / TPM



PKI & CRYPTO



CI/CD SECURITY

KEY DELIVERABLES



Cyber Security Plan
& Policies



TARA Report &
Risk Assessment



Security Requirements
& Architecture



Security Test Reports &
Vulnerability Reports



Compliance Pack
(ISO/SAE 21434)



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**“ENGINEERING EXCELLENCE TODAY,
BUILDING A SAFER TOMORROW.”**

THANK YOU!

FOR YOUR TIME. FOR YOUR **TRUST**. FOR YOUR **PARTNERSHIP**.

TOGETHER, WE ENGINEER **SAFETY**.
TOGETHER, WE DRIVE **IMPACT**.



SAFETY BY DESIGN

Building reliable, secure
and compliant solutions.



IMPACT DRIVEN

Delivering measurable
business value.



PARTNERSHIP LED

Collaborating closely for
shared success.



INNOVATION FOCUSED

Creating solutions that
anticipate the future.



GROWTH TOGETHER

Scaling new heights
with excellence.



We don't just build products,
we build confidence for millions of lives.



LET'S BUILD WHAT'S NEXT.
TOGETHER.



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